Free Energy Landscape Perturbation for Estimating Mutation-Induced Population Shifts from Reference Molecular Dynamics Ensembles

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Abstract

Protein function is governed by an ensemble of conformational states, and mutations can alter function by redistributing probability across that ensemble even when structural changes are modest. Directly estimating a perturbed conformational free-energy landscape with molecular dynamics (MD) is expensive because each mutation normally requires a new converged simulation. Here, we introduce Free Energy Landscape Perturbation (FELP), a state-local reweighting framework that estimates mutation-induced population shifts from a reference MD ensemble. FELP partitions a reference trajectory into discrete microstates, evaluates restrained reference-to-perturbed energy differences for configurations sampled within each state, and combines state-local Zwanzig factors with the reference state probabilities to estimate the perturbed state distribution. FELP estimates how a perturbation changes the relative probabilities of conformational basins and reports effective-sample and overlap diagnostics that identify when the predicted shift lacks support in the sampled reference ensemble. We describe the FELP estimator, its implementation, and planned validation on glycine dipeptide, chignolin, and T4 lysozyme. FELP is intended for perturbations that reshape populations within a sufficiently sampled reference landscape, providing a practical bridge between local alchemical energy evaluation and ensemble-level mechanistic interpretation.

Introduction

Protein function is determined by a free-energy landscape rather than by a single static structure. Folding, stability, enzyme catalysis, signal transduction, molecular recognition, and allosteric regulation all depend on how probability is distributed over conformational states and how readily the system moves between them. Many proteins therefore function through ensembles of native-like substates, with activity controlled by changes in basin depth, transition pathways, and relative occupancy. Quantitatively describing these landscapes is central to understanding how sequence, ligand binding, protonation, and chemical modification alter biological function [1].

Molecular dynamics (MD) simulations provide a direct route to estimating these conformational free-energy landscapes at atomistic resolution. In principle, sufficiently long trajectories recover both the relevant structural diversity and the thermodynamic probability of each state. In practice, sampling remains the central limitation. Transitions between functional conformations can be slow, rare, and system-specific, so converging a separate MD ensemble for every perturbation of interest is often prohibitively expensive. Parallel simulation strategies, enhanced sampling, and Markov state models (MSMs) have improved access to long-timescale behavior by combining many trajectories into kinetic and thermodynamic state models [1, 2]. Effective sample size and related convergence diagnostics provide complementary measures of whether a simulation has generated independent structural information, rather than simply accumulated correlated frames [3, 4]. Even with these advances, each new mutation or chemical change can still demand substantial additional sampling.

Alchemical free-energy methods make a different tradeoff. Free energy perturbation (FEP) and related

estimators compute free-energy differences between Hamiltonians from equilibrium samples and are formally rigorous when phase-space overlap is sufficient [5–10]. Phase-space overlap controls both bias and reliability in exponential averaging, Bennett acceptance ratio, thermodynamic integration, and staged alchemical calculations [11, 12]. These approaches are powerful for a defined thermodynamic cycle, and practical calculations are often local: they are initialized from one or a small number of conformations and restrained to prevent structural drift. Such calculations can estimate the energetic effect of a mutation or ligand in a particular structural context. Additional ensemble analysis is required to describe how that perturbation redistributes probability across a conformational landscape. Because exponential estimators are sensitive to rare high-weight samples, inadequate overlap can also produce unstable estimates that look numerically complete while remaining physically unsupported.

Single-structure and local relaxation methods face the complementary limitation. Rosetta-style mutation stability calculations, for example, can test whether a particular coordinate set becomes more or less favorable after mutation, but they typically evaluate a narrow structural hypothesis [13]. Minimal relaxation avoids the cost of full resampling, yet it can convert small coordinate differences into abrupt energy changes and noisy estimates. These calculations are useful for local stability screening, but they do not directly answer whether a mutation stabilizes one conformational basin over another.

Recent population-shift strategies address the ensemble problem by separating conformational sampling from perturbation evaluation. Implicit-ligand theory and related alchemical-state formulations showed how binding or perturbation thermodynamics can be estimated from conformational ensembles by reweighting state probabilities or thermodynamic expectations [14–17]. PopShift applies this idea to ligand binding by estimating how ligand interactions change the probabilities of conformational states sampled in a ligand-free MSM [18]. The key insight is that expensive sampling can be performed once for a reference system, after which many perturbations can be evaluated against the same state model. This framework moves beyond single-conformation estimates by explicitly linking perturbation effects to the thermodynamic weight of each conformational state.

Here, we introduce Free Energy Landscape Perturbation (FELP), a mutation-focused generalization of this idea that combines reference MD sampling, discrete state models, restrained local perturbation evaluation, and FEP-style reweighting. FELP starts from a reference ensemble, partitions sampled configurations into microstates in a chosen feature space, evaluates the energetic effect of a mutation within each microstate, and reweights the reference state probabilities to estimate the perturbed landscape. The method treats a global population shift as a collection of local, tractable reweighting problems. Critically, FELP also audits the shift using effective-sample-size, weight-concentration, and discrete-overlap diagnostics, making explicit when a proposed perturbation is outside the regime supported by the reference ensemble.

Theory

Consider a reference system with potential energy $U_0(x)$ and a perturbed system with potential energy $U_1(x)$, where x denotes a molecular configuration and $\beta = (k_B T)^{-1}$. The Boltzmann distribution maps potential energy functions to probability densities over configurations. A reference trajectory samples configurations from the reference density

$$p_0(x) = \frac{\exp[-\beta U_0(x)]}{Z_0}. \tag{1}$$

The quantity $p_0(x)$ is the continuous probability density of configuration x under the reference Hamiltonian. FELP partitions this configuration space into discrete microstates $i \in \{1, \ldots, N\}$ using a chosen collective-variable or feature representation. Let $\chi_i(x)$ be the indicator function for state i. The restricted partition function for state i under Hamiltonian a is

$$Z_a(i) = \int \chi_i(x) \exp[-\beta U_a(x)]\, dx. \tag{2}$$

The corresponding microstate probability is

$$\pi_a(i) = \frac{Z_a(i)}{\sum_k Z_a(k)}. \tag{3}$$

For the reference system, this is equivalent to integrating the continuous reference density over the microstate: $\pi_0(i) = \int \chi_i(x) p_0(x)\, dx$.

The perturbed system has its own Boltzmann density,

$$p_1(x) = \frac{\exp[-\beta U_1(x)]}{Z_1}, \tag{4}$$

and the desired perturbed probability of the same conformational microstate is $\pi_1(i) = Z_1(i)/\sum_k Z_1(k)$. FELP estimates this quantity from reference samples assigned to the same microstate.

Within a microstate, the ratio of restricted partition functions follows the two-state Zwanzig identity,

$$\frac{Z_1(i)}{Z_0(i)} = \langle \exp[-\beta \Delta U(x)] \rangle_{0,i}\,, \tag{5}$$

where $\Delta U(x) = U_1(x) - U_0(x)$ and $\langle \cdot \rangle_{0,i}$ denotes an average over reference configurations conditioned on membership in state i. Normalizing the restricted partition functions over all microstates gives

$$\pi_1(i) = \frac{\pi_0(i)\, \langle \exp[-\beta \Delta U(x)] \rangle_{0,i}}{\sum_k \pi_0(k)\, \langle \exp[-\beta \Delta U(x)] \rangle_{0,k}}, \tag{6}$$

Equation 6 is the central FELP estimator. It preserves the reference state decomposition and updates the probability of each state according to the perturbation's energetic effect on configurations sampled in that state.

This is a two-state Zwanzig construction applied repeatedly across conformational microstates. The two thermodynamic states are the reference Hamiltonian U_0 and the perturbed Hamiltonian U_1. The microstates i define conformational conditioning regions where local phase-space overlap is expected to be stronger than overlap across the full ensemble. Shared membership in a microstate is therefore the operational overlap assumption in FELP: reference configurations assigned to state i must represent the configurations that carry meaningful perturbed probability within that same state. A multistate estimator such as MBAR would be appropriate if samples were available from multiple alchemical or thermodynamic states, such as intermediate λ windows or a directly sampled perturbed ensemble. FELP addresses the reference-sampling regime, where expensive sampling is available primarily for the reference system and the key question is whether each microstate has enough local overlap to support the restricted Zwanzig estimate.

In a finite trajectory, state i contains evaluated configurations x_{ij}, $j = 1, \ldots, n_i$. FELP computes

$$f_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \exp[-\beta(\Delta U_{ij} - s)], \tag{7}$$

where $s = \min(\Delta U)$ over all eligible evaluated frames is a numerical stability shift. The shift cancels during normalization, so the implemented estimator remains

$$\hat{\pi}_1(i) = \frac{\hat{\pi}_0(i) f_i}{\sum_k \hat{\pi}_0(k) f_k}. \tag{8}$$

The corresponding relative free energy is

$$F_1(i) = -k_B T \ln \hat{\pi}_1(i) + C, \tag{9}$$

where C is chosen so that the lowest finite state free energy is zero.

The central hypothesis is that discretization creates local overlap domains: within each microstate, reference configurations and perturbed configurations share enough phase space for the two-state Zwanzig factor in Equation 5 to estimate the change in that state's restricted partition function. Under this hypothesis, the global population shift can be decomposed into state-local two-Hamiltonian reweighting problems. If a mutation is unfavorable in one basin but favorable in another, FELP predicts a population shift between

basins rather than a single global stability score. The estimate is conditional on the state decomposition, reference prior, restraint protocol, and energy model used for the local comparisons.

This estimator is meaningful only when the perturbed ensemble is represented within the reference support of each state that contributes appreciably to π_1. If important perturbed configurations were not sampled within their assigned reference microstates, no reweighting procedure can recover them. FELP therefore treats phase-space support as part of the theory rather than as a post hoc plotting concern. Importance-sampling theory motivates the use of normalized weights and effective-sample diagnostics to measure weight degeneracy [19, 20]. For state i, the effective sample size is

$$\mathrm{ESS}_i = \frac{\left(\sum_j w_{ij}\right)^2}{\sum_j w_{ij}^2}, \tag{10}$$

where $w_{ij} = \exp[-\beta(\Delta U_{ij} - s)]$. Low ESS_i indicates that the state-local perturbation factor is dominated by a small number of favorable frames rather than by broad support within the state.

FELP also computes global weight-concentration diagnostics from the normalized frame weights across all eligible frames. These include the global effective-sample fraction, the maximum normalized frame weight, and the probability mass carried by the top 1%, 5%, and 10% of frame weights. To quantify support at the state-distribution level, FELP reports the target probability mass assigned to low-support states, such as states with $\mathrm{ESS} < 10$ or $\mathrm{ESS} < 50$, and a discrete overlap coefficient

$$\Omega = \sum_i \min[\hat{\pi}_0(i), \hat{\pi}_1(i)]. \tag{11}$$

The coefficient Ω measures overlap between the reference and reweighted discrete state distributions on the sampled state space. This diagnostic identifies large departures from the sampled reference prior, while leaving open the possibility of target-only basins outside the reference trajectory.

The current implementation labels support as "ok," "caution," or "poor" using advisory thresholds. Poor support is flagged when the global effective-sample fraction is below 0.01 or when more than 25% of target probability lies in states with $\mathrm{ESS} < 10$. Caution is flagged when the global effective-sample fraction is below 0.05, when more than 10% of target probability lies in states with $\mathrm{ESS} < 10$, or when $\Omega < 0.20$. These practical thresholds warn that the reweighted distribution may be dominated by limited reference support. These diagnostics define the regime in which the FELP estimate should be trusted.

Workflow

The FELP workflow has four stages **(Figure 1)**. First, a reference MD trajectory is generated for the system expected to contain the broader or more permissive conformational ensemble. The trajectory is summarized into microstates using a uniform ϕ/ψ grid, an explicit JSON/YAML state file, automatically generated two-dimensional bins, or an exported MSM bundle. These microstates define both the aggregation units for reweighting and the coordinates used for plotting.

Second, each reference frame is assigned to a microstate. When raw-count priors are used, $\hat{\pi}_0(i)$ is the empirical fraction of assigned reference frames in state i. When a state file or MSM bundle contains trusted reference probabilities, FELP can instead normalize those loaded priors. The state identifiers used during assignment become the identifiers used throughout reweighting, summary tables, and plotting.

Third, FELP evaluates selected configurations within each state. Amino acid substitutions are introduced with PyRosetta. Reference-frame restraints are built from the original configuration and then applied to both the reference and perturbed systems so that the energy comparison remains local to the assigned state. The current implementation supports Rosetta preparation by FastRelax or side-chain repacking, target-only or radius-based mobility, and energy evaluation with OpenMM/AMBER or Rosetta scoring. The output of this stage is a frame-level table containing U_0, U_1, ΔU, assignment metadata, restraint deviations, and collective-variable measurements.

Fourth, FELP aggregates frame-level Boltzmann factors into state-level reweighting factors and computes the perturbed state probabilities. The resulting landscape can be plotted over discrete microstates or projected into the collective-variable space used for assignment. For custom collective variables, FELP stores state ranges, state centers, and frame-level measurements as metadata so that downstream visualization remains tied to the same coordinates used for reweighting.

Placeholder for FELP workflow schematic.

Figure 1: Free Energy Landscape Perturbation workflow. A reference MD ensemble is partitioned into microstates in a chosen collective-variable space. Reference-to-perturbed energy differences are evaluated for sampled structures within each state. State-specific exponential averages reweight the reference probabilities to estimate the perturbed landscape. Overlap diagnostics audit whether the predicted population shift is supported by the reference ensemble.

Automatic microstate construction for custom collective variables

For simple Ramachandran analyses, a uniform ϕ/ψ grid is a natural microstate definition. For custom distance coordinates, fixed generic bins can be inefficient: overly fine bins produce unsupported states and noisy reweighting factors, while overly coarse bins obscure landscape features. FELP therefore supports automatic microstate generation for two-dimensional custom collective-variable spaces. When two collective variables are supplied without an explicit microstate file, FELP scans the reference trajectory, measures the configured variables, chooses conservative bins from the observed marginal and joint statistics, and writes a reusable microstate JSON file.

The automatic binning procedure is designed for both reweighting and visualization. Marginal bin counts are initialized using the Freedman–Diaconis rule, which adapts to the spread and sample size of the observed distribution. The joint grid is then capped by an expected frames-per-state criterion so that the state decomposition remains statistically supported. The generated file contains a complete rectangular grid with explicit "cv_ranges," state identifiers, plot coordinates, observed ranges, quantiles, bin edges, and policy metadata. Subsequent runs can reuse this file directly, ensuring that comparisons across mutations use identical microstate definitions.

Benchmark design

To evaluate FELP, we will compare reweighted perturbed landscapes against conventional MD simulations of the corresponding perturbed systems. The benchmark design is intentionally asymmetric: the reference simulation should cover the larger or more permissive conformational space, while the perturbation should redistribute probability within that sampled space. This design tests the intended use case of FELP. It does not ask reweighting to discover conformations absent from the reference ensemble.

For each system, we construct a reference landscape, introduce a perturbation, compute the FELP-reweighted landscape, and compare the result to an independently sampled perturbed landscape. Agreement will be assessed at multiple levels: qualitative landscape topology, state population shifts, free-energy differences between dominant basins, and overlap diagnostics. Cases with poor overlap will be treated as informative failures rather than hidden errors.

Glycine dipeptide

Glycine dipeptide provides a minimal test of FELP in a well-understood two-dimensional conformational landscape. The reference ensemble is projected into backbone Ramachandran space and partitioned into

ϕ/ψ microstates. The current benchmark compares a glycine reference ensemble to an alanine-like perturbed ensemble, a useful first case because glycine samples a broader Ramachandran landscape than alanine. The expected result is recovery of the dominant population redistribution rather than every high-energy contour: alanine should down-weight regions that are favorable only for glycine and retain probability in the subset of Ramachandran space compatible with the perturbed side chain.

This benchmark is also the clearest setting for diagnosing the distinction between phase-space support and energetic preference. If a Ramachandran basin exists in the alanine MD reference but was absent from the glycine-to-alanine FELP support, FELP should not be expected to recover it. Conversely, if the alanine landscape is contained within the glycine support, FELP should reproduce the major basin-level shifts with favorable effective-sample diagnostics. The glycine system will therefore serve as the primary validation of the estimator and plotting semantics.

Placeholder for glycine dipeptide reference, MD-perturbed, and FELP-reweighted landscapes.

Figure 2: FELP benchmark on a Ramachandran landscape. TODO: show the reference landscape, the independently sampled perturbed landscape, the FELP-reweighted landscape, and the difference map. Include overlap diagnostics and basin population comparisons.

Chignolin

Chignolin tests FELP on a small protein whose conformational preferences depend on coordinated intramolecular contacts. Rather than projecting only the target residue's backbone angles, we define collective variables based on intramolecular distances that report on global compaction and turn geometry. This benchmark is intended to test whether FELP can predict mutation-induced shifts in a more protein-like landscape while retaining a compact and interpretable state representation.

In this setting, naive use of a default Ramachandran grid would be misleading: the prior would be defined over target-residue ϕ/ψ bins rather than the distance coordinates used to interpret the landscape. FELP therefore treats the distance variables as first-class collective variables. The reference trajectory is binned in the joint distance space, raw-count priors are computed over those distance-defined microstates, and the perturbed landscape is visualized in the same coordinates.

The expected chignolin result is a mutation-dependent redistribution between compact, turn-like, and extended-like regions of the distance-coordinate landscape. A successful FELP run should preserve the broad topology of the independently sampled mutant landscape while identifying which microstates gain or lose probability. The analysis should report both the continuous projected surfaces and the discrete state summaries, because discrepancies between these views can reveal whether a visually smooth projected landscape is supported by enough state-local samples.

T4 lysozyme

T4 lysozyme provides a natural larger benchmark because it has been used extensively to study ligand binding, cavity mutants, and population-shift methods, including the PopShift validation on L99A [18, 21]. For FELP, the T4 lysozyme benchmark should be framed around mutations or chemical perturbations that redistribute probability among pre-existing conformational substates rather than requiring discovery of an entirely new fold or pocket geometry.

Placeholder for chignolin distance-CV landscapes.

Figure 3: Distance-coordinate FELP benchmark in chignolin. TODO: show the reference distance landscape, independently sampled mutant landscape, FELP-reweighted landscape, and population shifts for folded, turn, and extended-like states. Report the automatically generated binning or the explicit microstate JSON used for the analysis.

In this setting, the state definition should be chosen to reflect the mechanistic question. Possible collective variables include cavity-opening distances, helix or loop distances surrounding the engineered binding pocket, side-chain rotameric states of pocket residues, or MSM states derived from reference simulations. The expected result is a more stringent test than glycine or chignolin: FELP should identify mutation-induced shifts among protein substates when the reference trajectory samples those substates, while the diagnostics should flag cases in which the mutation stabilizes configurations absent from the reference simulation. This benchmark will therefore test both the scalability of FELP and the practical value of its overlap audit.

Overlap diagnostics and failure modes

FELP is not intended to replace perturbed MD when the perturbation creates a new basin absent from the reference simulation. Its value lies in rapidly estimating population shifts within a sampled landscape and in identifying when that estimate is unsupported. The most important failure mode is poor phase-space overlap. In practice, this appears as a small effective sample size, target probability concentrated in low-support states, one or a few frames dominating the reweighted distribution, or positive-prior states with no evaluated support.

The support diagnostics introduced in the Theory section distinguish FELP from purely empirical population-shift tools. A population shift can always be computed mathematically; physical credibility depends on reference support. By reporting overlap statistics alongside the reweighted landscape, FELP converts this into an auditable claim: the user can ask what shift is predicted and whether the reference ensemble contains the configurations required for that prediction to be meaningful. When both directions are computationally feasible, forward and reverse FELP calculations provide a stronger overlap test.

Discussion

FELP occupies an intermediate point between high-fidelity perturbed MD and local mutation scoring. It does not attempt to resample every mutant. Instead, it leverages extensive sampling of a reference system and asks how chemical perturbations reweight the probabilities of already observed states. This makes FELP most appropriate for perturbations expected to modulate existing conformational preferences rather than create entirely new modes of motion.

The method clarifies the relationship between physical rigor and computational throughput. The local estimator is inherited from FEP and therefore carries the same overlap requirements as any exponential reweighting calculation. The global landscape estimate is obtained by applying these local perturbation factors across a reference state model. When the reference state model is well supported and overlap diagnostics are favorable, FELP can estimate mutation-induced population shifts at far lower cost than independently converging each perturbed ensemble. When diagnostics are unfavorable, the method provides evidence that new sampling is required.

The most important conceptual limitation is that FELP estimates the perturbed distribution over the reference state space. It cannot discover a target-only state, and it should not be described as a replacement

for perturbed MD when large conformational changes are expected. The practical strength of the method is that this limitation is explicit. FELP makes a prediction and a validity statement together: the predicted shift, the state support for that shift, and the extent to which the estimate is dominated by rare high-weight frames.

Several implementation details are important for practical use. First, the collective variables used for microstate assignment must match the biological question and the plotted landscape. Second, state identifiers in the microstate definition are the identifiers used for raw-count priors, reweighting, and plotting. Third, custom collective variables must be propagated through frame metadata and summary metadata so that visualization does not silently fall back to default ϕ/ψ coordinates. Finally, restraint strengths and coordinate-preservation protocols must be interpreted in the units and conventions of the engine being used. In the current implementation, Rosetta and OpenMM use different restraint parameterizations, so backend-specific restraint settings must be reported in Methods.

The planned benchmarks are designed to expose these strengths and limitations. Glycine dipeptide tests whether the estimator behaves correctly in a small, visually interpretable landscape. Chignolin tests whether distance-coordinate microstates can capture protein-like population shifts. T4 lysozyme tests whether FELP can scale to a biologically studied system while providing meaningful warnings when a mutation or ligand-coupled perturbation moves beyond the reference support. Together, these systems should establish FELP as an auditable reweighting framework for a defined and useful class of perturbations.

Conclusion

Free Energy Landscape Perturbation provides a framework for estimating how mutations and other chemical perturbations reshape conformational free-energy landscapes without requiring a new converged MD simulation for every perturbed system. By combining reference ensemble sampling, microstate-based reweighting, local energy-difference evaluation, and overlap diagnostics, FELP connects the efficiency of perturbative calculations with the ensemble perspective required for protein function. The resulting approach is most powerful when the reference simulation captures the relevant conformational space and when diagnostics confirm that the predicted population shift is supported. Under these conditions, expansive sampling performed once for a reference model can be reused to test a family of perturbations while preserving a direct connection to thermodynamic state populations.

Methods

Reference simulations and trajectory processing

Reference trajectories are supplied to FELP as one or more topology/trajectory groups. Files passed within one trajectory group are treated as contiguous fragments of the same simulation, whereas repeated trajectory groups are treated as independent trajectories for MSM construction and provenance. For each benchmark, the final manuscript should report the MD engine, force field, solvent model, temperature, pressure treatment, time step, thermostat and barostat settings, equilibration protocol, production length, number of independent replicas, and frame stride used for FELP analysis.

Trajectory frames are loaded with a user-specified topology, optional frame start/stop indices, and stride. For each selected frame, FELP clones the reference topology with the frame coordinates, converts the structure to a PyRosetta pose when needed, and records the trajectory identifier and source frame index. These identifiers are propagated to the frame-level output so that failed frames, high-weight frames, and representative microstate samples can be traced back to the original simulation data.

Microstate construction

For Ramachandran analyses, microstates are defined on a uniform ϕ/ψ grid over $[-180°, 180°)$ for each axis. For custom collective-variable analyses, microstates are defined by explicit JSON or YAML files containing state identifiers and "cv_ranges." State definitions may include plotting coordinates and loaded reference probabilities. If no probabilities are provided, FELP can compute raw-count priors from the assigned reference

trajectory. Angular intervals are treated circularly so that states crossing the $-180°/180°$ boundary are handled correctly.

For two-dimensional custom collective variables, FELP can automatically construct a reusable microstate file. The trajectory is scanned, configured backbone dihedrals and/or distance CVs are measured, marginal bin counts are initialized using the Freedman–Diaconis rule, and the joint grid is capped by a support criterion targeting approximately 25 frames per state with a maximum of 144 total states. The generated file records bin edges, observed ranges, quantiles, state centers, and policy metadata. All subsequent reference and perturbed analyses for a benchmark should use the same microstate definitions.

MSM-backed analyses use a previously exported PyEMMA bundle. The bundle supplies precomputed assignments, cluster or metastable state definitions, and optionally loaded stationary probabilities. The FELP reweighting equation is unchanged; the MSM provides the state space and reference prior.

Perturbation and energy evaluation

Mutations are introduced with PyRosetta at a user-specified sequential pose residue index. FELP optionally enforces the original residue identity before scoring, which is useful for glycine-to-alanine or protonation-state studies where the reference topology must be chemically explicit. Canonical amino acids and common protonation or disulfide variants are supported when the corresponding residue types are available to PyRosetta and the selected energy backend.

For each frame, FELP first measures the configured collective variables and assigns the frame to a microstate. It then builds reference-frame restraints from the same pose. Default restraints are the target residue ϕ and ψ angles, but users may supply arbitrary backbone dihedral CVs and distance CVs. The same restraint targets are applied to the reference and perturbed systems. This makes the comparison local to the assigned reference configuration and reduces drift into a different basin before the energy difference is evaluated.

The PyRosetta preparation step can use FastRelax or side-chain repacking. The default mobility policy allows the target residue side chain to move while keeping the rest of the structure fixed; a radius-based mobility option can permit neighboring side chains, and optionally backbone atoms, to relax. For Rosetta preparation, the default score function is ref2015 with Cartesian relaxation enabled when supported. Backbone CVs are explicitly reset and checked after mutation and local preparation. Distance restraints are implemented as Rosetta atom-pair constraints with a reported width parameter, whereas OpenMM distance restraints use harmonic spring constants; both settings should be reported for production analyses.

Energy evaluation can be performed with OpenMM or Rosetta. The default OpenMM backend uses AMBER ff14SB-only side-chain protein parameters with GBNeck2 implicit solvent, evaluates both reference and perturbed structures under the same restraint targets, freezes atoms outside the configured mobility region during minimization, and reports energies in $\mathrm{kJ\,mol^{-1}}$. The optional Rosetta backend scores PyRosetta poses with the configured score function and constraints. In either backend, FELP records

$$\Delta U_{ij} = U_1(x_{ij}) - U_0(x_{ij}) \tag{12}$$

for each evaluated frame and excludes failed or non-finite frames from reweighting.

Reweighting and diagnostics

State probabilities are reweighted according to Equation 6. The reference prior is either loaded from the microstate definitions or estimated from raw assigned frame counts. When microstate-stratified sampling is used to reduce expensive energy evaluations, FELP assigns all frames cheaply, stores a deterministic sampling manifest, computes raw-count priors from the full assigned support, and estimates state-local Boltzmann averages from uniformly selected frames within each state. The default evaluates all assigned frames.

Diagnostics include global and state-level effective sample size, maximum normalized frame weight, top-weight mass fractions, overlap coefficient, unsupported positive-prior states, and target probability carried by low-effective-sample-size states. The current advisory thresholds are described above and written to the run summary. Final analyses should report these diagnostics alongside each reweighted landscape rather than presenting the landscape as an unconditional prediction.

Visualization and comparison to perturbed MD

FELP writes frame-level results, microstate summaries, reweighted surfaces, run configuration, sampling manifests, and JSON summaries. Discrete microstate plots use the state probabilities and relative free energies from the microstate summary. Projected free-energy plots use state-consistent frame weights to construct weighted histograms in the plotted collective-variable coordinates. Free energies are shifted so that the lowest finite value is zero; only differences between states or plotted bins are meaningful.

For each benchmark, the FELP-reweighted landscape will be compared to an independently sampled perturbed MD landscape. Planned comparisons include the identity and location of dominant basins, relative free energies between basins, per-state occupancy changes, target-versus-reference probability scatter plots, KL-divergence or occupancy-shift summaries, and whether states with the largest predicted changes are supported by adequate effective sample sizes.

Code and data availability

The FELP implementation, analysis scripts, run commands, generated microstate definitions, processed summaries, and plotting inputs will be released in a public repository and archived in a DOI-bearing repository before submission.

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Author contributions

T.E.M. and B.P.B. conceived the study. T.E.M. performed simulations and analyses. B.P.B. developed the FELP software framework. T.E.M. and B.P.B. interpreted the results and wrote the manuscript.

Disclosures

The authors declare no competing financial interest.

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